NAME: KARTHIK R NAYAK
SRN: PES2UG24CS217

SECTION D

5TH SEM

REQUIREMENTS

FUNCTIONAL REQUIREMENTS

FR-001

DESCRIPTION
The system will use GPS metadata from the uploaded pictures of damage in order to determine the location of the incident automatically on the GIS map of the municipality.

PRIORITY : HIGH

ACCEPTANCE CRITERIA
Pass: A geotagged picture leads to the location of the incident being automatically shown on the GIS map.

Fail: The location of the incident cannot be automatically identified without GPS metadata.

Automates location identification of the incident.

RATIONALE

Automates incident location identification.

FR-002

DESCRIPTION

The system shall enable a Citizen Reporter to make reports on road potholes, broken streetlights, and water pipeline leakage.

PRIORITY HIGH

ACCEPTANCE CRITERIA

Pass: The citizen selects any of the three categories of damages and submits the report.

Fail: The system does not accept a submission of the report without selection of a category of damage.

RATIONALE

Offers the core function for submitting reports about infrastructure damages.

FR-003

DESCRIPTION

The system shall enable a Citizen Reporter to attach a photograph of the damage he reports.

PRIORITY HIGH

ACCEPTANCE CRITERIA

Pass: A valid photograph is attached with the report.

Fail: The system does not attach the uploaded photograph to the report.

RATIONALE

Provides physical proof of the damage being reported.

FR-004

DESCRIPTION

The system shall automatically assign the damage reports to the concerned municipal engineering departments.

PRIORITY HIGH

ACCEPTANCE CRITERIA

Pass: A pothole damage report is assigned to the Road Infrastructure department.

Fail: The damage report is not assigned or is assigned to the wrong department.

Ensures assignment of reports to the right municipal department.

FR-005

DESCRIPTION

The system shall allow a Municipal Engineer to review the assigned damage reports.

PRIORITY : HIGH

ACCEPTANCE CRITERIA

Pass: The engineer can view the assigned damage reports and its details.

Fail: The engineer cannot access the assigned report.

RATIONALE

Enables municipal engineers to act on citizen reports.

NONFUNCTIONAL REQUIREMENTS

NFR-001

DESCRIPTION

The reporting portal will have the capability of offline report preparation and synchronization once the mobile network connectivity is re-established.

PRIORITY: High

ACCEPTANCE CRITERIA

Pass: Offline prepared report will be saved and then synchronized once the network connectivity is back to normal.

Fail: Report prepared offline gets lost or not synchronized. Enables reporting regardless of network disconnection.

NFR-002

DESCRIPTION

The reporting portal will have the ability of maintaining the specified latency and security during simulated peak loads.

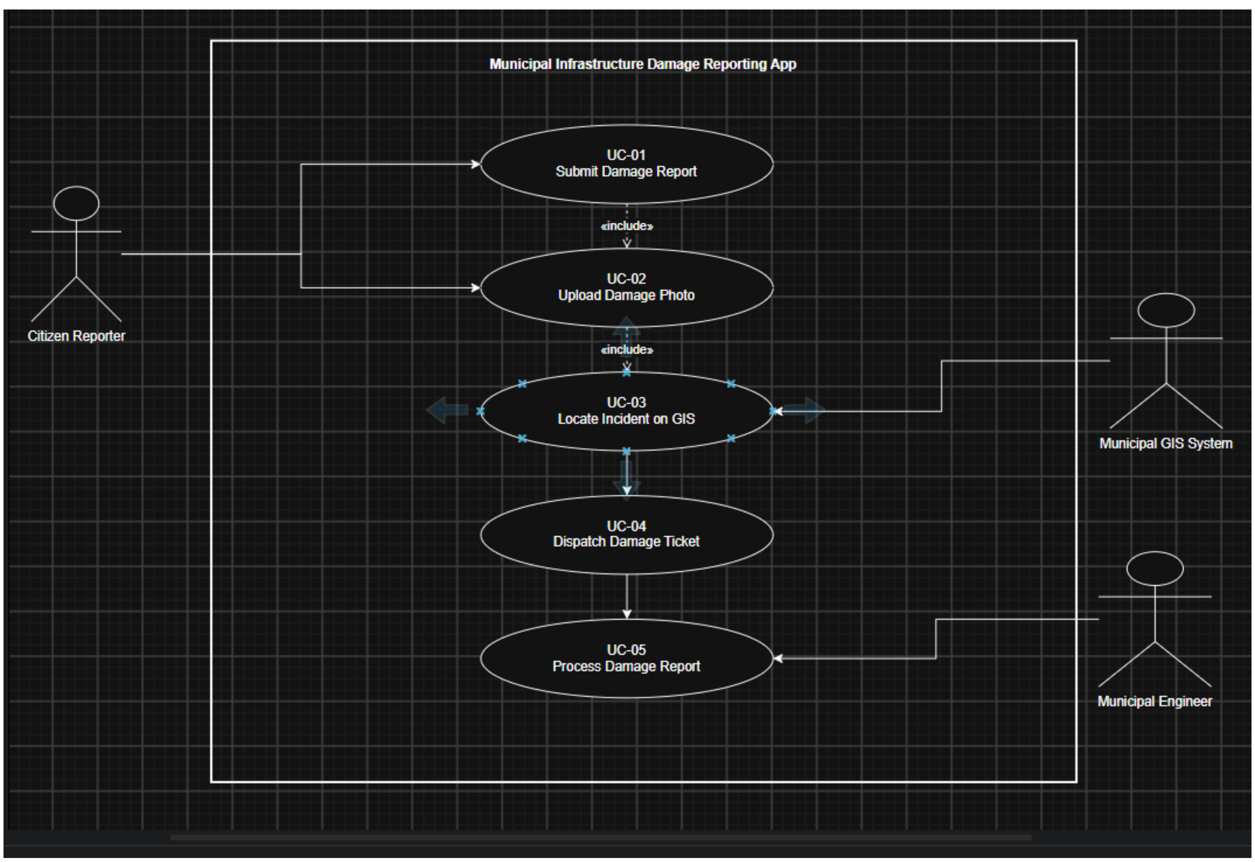
PRIORITY: High

Pass: The reporting portal passes benchmark tests with regard to latency and security during simulated peak loads.

Fail: The reporting portal does not pass either of the tests.

RATIONALE

Makes sure that the reporting portal is still responsive and secure at peak times.



USE CASE FLOW

Use Case ID: UC-01 Primary Actor: Citizen Reporter

1. Preconditions

1 The Citizen Reporter has access to the Municipal Infrastructure Damage Reporting App.
2 The citizen has identified a road pothole, broken streetlight, or water pipeline leak.
3 The citizen has a photograph of the reported damage available for upload.

2. Postconditions

1 A damage report is created in the system.
2 The uploaded photograph is associated with the report.
3 The incident location is identified using GPS information when available.
4 The damage ticket is dispatched to the relevant municipal engineering department.

3. Main Success Scenario

1 The Citizen Reporter opens the Municipal Infrastructure Damage Reporting App.
2 The citizen selects the type of infrastructure damage: pothole, broken streetlight, or water pipeline leak.
3 The citizen enters the required report information.
4 The citizen uploads a photograph of the reported damage.
5 The system extracts EXIF GPS metadata from the uploaded photograph.
6 The system uses the GPS coordinates to pinpoint the incident on the municipal GIS map.
7 The system creates a damage ticket containing the report details, photograph, and location.
8 The system automatically dispatches the ticket to the relevant municipal engineering department.

9 The system confirms successful submission of the report to the Citizen Reporter.
10 The use case ends successfully.

4. Alternate Flow — GPS Metadata Unavailable

1 5a1. The system cannot extract GPS coordinates from the uploaded photograph.
2 5a2. The system informs the Citizen Reporter that location information is unavailable.
3 5a3. The citizen provides the required location information through the available reporting mechanism.
4 5a4. The system uses the provided location to identify the incident location.
5 5a5. The system continues from Step 7 of th